

CampusNova University ? Smart Fleet & Transport Operations Manual

1. Fleet Configuration & Capacity Guidelines

CampusNova operates a dedicated transit fleet serving day-scholar students across metropolitan pickup corridors:

- Fleet Vehicle Bus-01: 45 passenger seating capacity (North-East Corridor)
- Fleet Vehicle Bus-02: 40 passenger seating capacity (South Corridor)
- Fleet Vehicle Van-01: 15 passenger seating capacity (Special Shuttle & Express Corridor)

2. Route Optimization & TSP Sequencing

Pickup clusters are algorithmically formed using spatial KMeans clustering partitioned by vehicle capacity. Within each cluster, the vehicle route is sequenced via Traveling Salesperson Problem (TSP) optimization with Open Source Routing Machine (OSRM) street network matrices.

3. Boarding Pass & Safety Protocols

Students must present their digital QR transport pass upon boarding. Vehicles must not exceed 100% rated passenger capacity. Real-time GPS telematics track fleet speed, ETA updates, and schedule compliance.